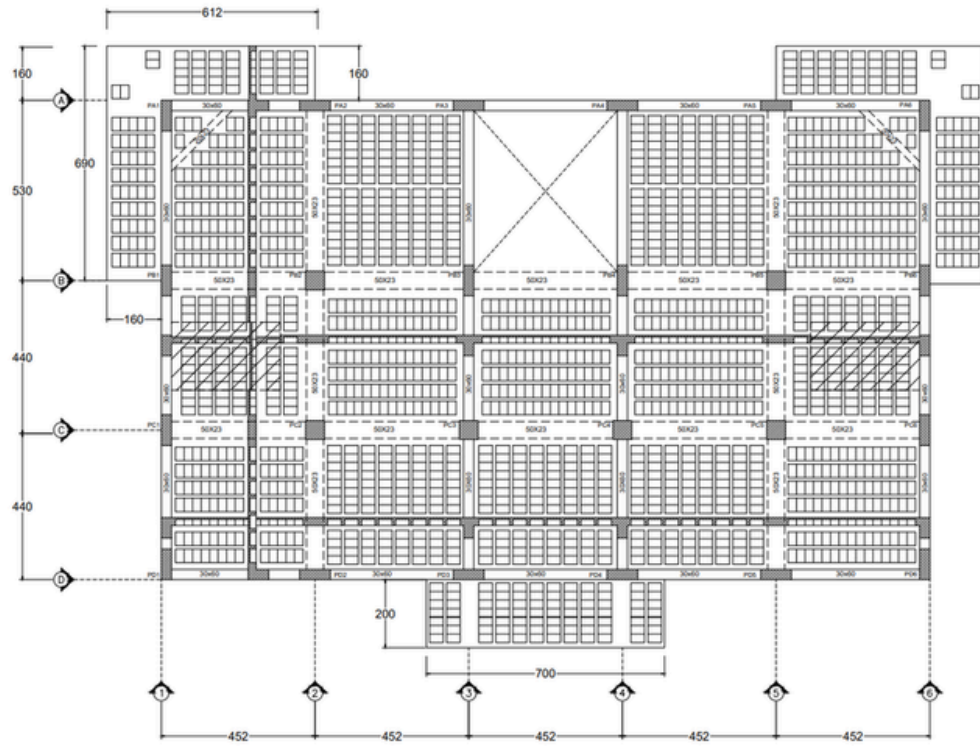


8-STOREY BUILDING STRUCTURAL DESIGN



TYPE: Structural Design

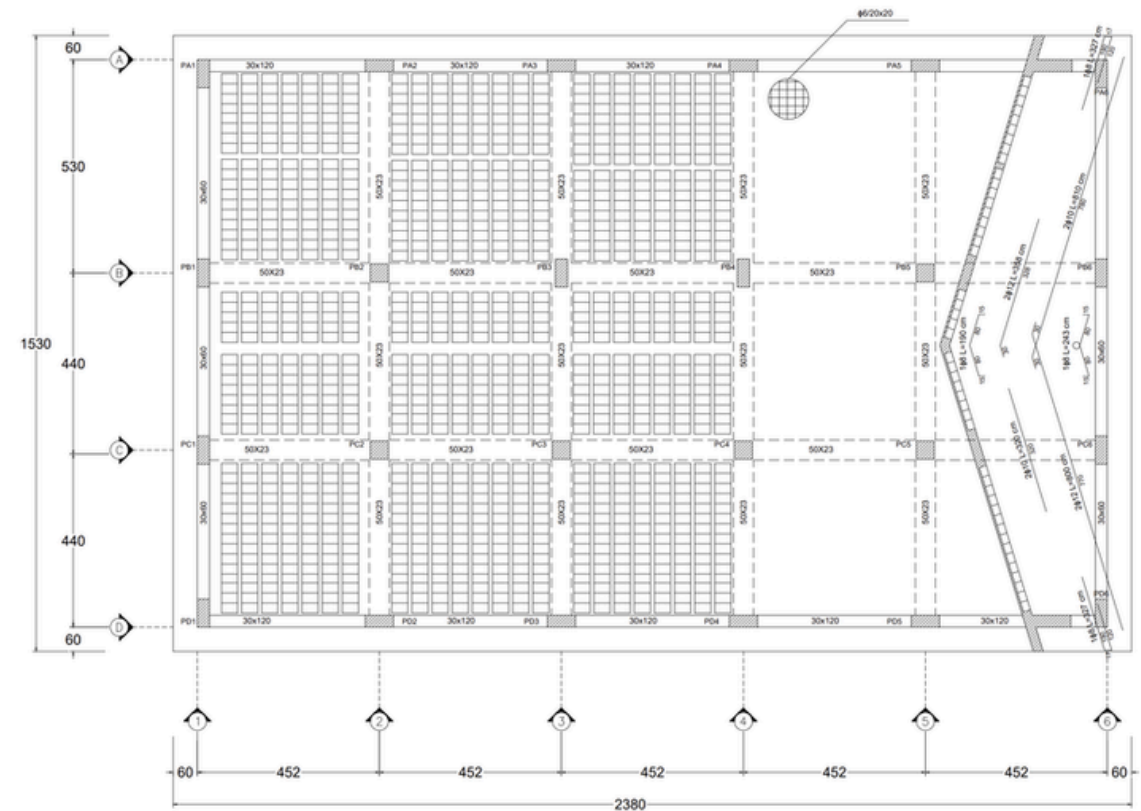
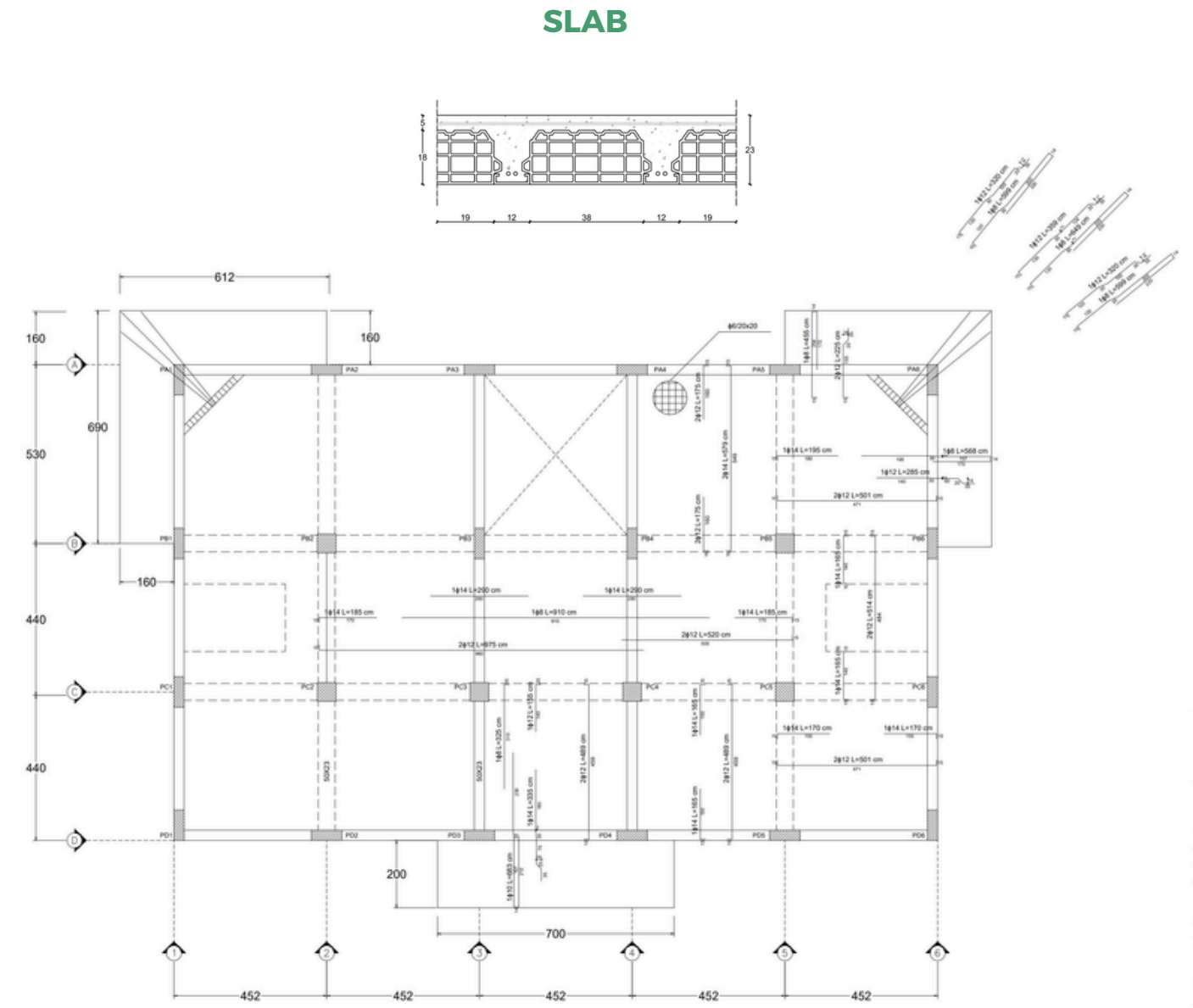
YEAR: 2023

TEAM: Andrea Cutroni & Alessandro Abagnale

FACULTY: Donato Abruzzese

CONTEXT: Academic course
“Special structures”

The project involves the structural design of an 8-storey building for residential and commercial use with a framed structure in reinforced concrete. The work includes the calculation with static and seismic verifications, according to Italian standards, of slabs, beams, columns, stairs and foundations. The project was designed using AutoCAD, SAP2000 and Excel.



The image displays structural design drawings for a bridge deck, including cross-sections and reinforcement details.

Top Section: Shows the bridge deck cross-section with reinforcement bars (red and blue) and a green shaded area representing the deck profile.

Bottom Section: Shows the reinforcement details for the bridge deck, including reinforcement bars (red and blue) and a green shaded area representing the deck profile.

Reinforcement Details:

- Top Reinforcement:**
 - 1Φ24 L=864 cm Pos 3
 - 1Φ22 L=874 cm Pos 2
 - 1Φ22 L=702 cm Pos 2
 - 2Φ18 L=753 cm Pos 1
 - 2Φ18 L=988 cm Pos 1
 - 1Φ16 L=728 cm Pos 4
 - 1Φ16 L=967 cm Pos 4
 - 1Φ10 L=356 cm
 - 2Φ14 L=995 cm Pos 5
 - 2Φ14 L=586 cm Pos 5
 - 1Φ22 L=823 cm Pos 6
 - 1Φ22 L=1115 cm Pos 6
 - 1Φ22 L=702 cm Pos 2
 - 2Φ18 L=753 cm Pos 1
 - 1Φ16 L=728 cm Pos 4
 - 1Φ10 L=356 cm
 - 2Φ14 L=995 cm Pos 5
 - 1Φ22 L=823 cm Pos 6
- Bottom Reinforcement:**
 - 1Φ24 L=864 cm Pos 3
 - 1Φ22 L=874 cm Pos 2
 - 1Φ22 L=702 cm Pos 2
 - 2Φ18 L=753 cm Pos 1
 - 2Φ18 L=988 cm Pos 1
 - 1Φ16 L=728 cm Pos 4
 - 1Φ16 L=967 cm Pos 4
 - 1Φ10 L=356 cm
 - 2Φ14 L=995 cm Pos 5
 - 2Φ14 L=586 cm Pos 5
 - 1Φ22 L=823 cm Pos 6
 - 1Φ22 L=1115 cm Pos 6
 - 1Φ22 L=702 cm Pos 2
 - 2Φ18 L=753 cm Pos 1
 - 1Φ16 L=728 cm Pos 4
 - 1Φ10 L=356 cm
 - 2Φ14 L=995 cm Pos 5
 - 1Φ22 L=823 cm Pos 6

Dimensions:

- Span length: 20m
- Deck width: 24m
- Reinforcement spacing: 150mm
- Reinforcement diameter: 10mm, 16mm, 18mm, 22mm, 24mm

The drawing illustrates the reinforcement details for a reinforced concrete column. The longitudinal section on the left shows the column's profile with varying diameters and the placement of reinforcement bars at specific levels. The cross-sectional views on the right provide a detailed look at the reinforcement layout for three different column diameters: 70 cm, 64 cm, and 90 cm. Each cross-section shows the arrangement of longitudinal bars (3Φ22, 4Φ22, 3Φ24, 6Φ24) and stirrups (Φ8).

Figure 1 illustrates the reinforcement details for the beam-column joints. The top diagram shows a 3D perspective of a beam-column joint with dimensions 2.85 m and 1.30 m. Below are five 2D reinforcement layout diagrams for different beam segments, each showing bar counts, lengths, and angles.

- Top Left:** 2 ϕ 10 L=230 cm. Dimensions: 100, 115, 100.
- Bottom Left:** 2 ϕ 12 L=210 cm. Dimensions: 100, 100, 45.
- Middle Left:** 2 ϕ 12 L=825 cm. Dimensions: 85, 300, 45.
- Middle Right:** 2 ϕ 12 L=295 cm. Dimensions: 45, 150, 100.
- Bottom Right:** 2 ϕ 12 L=370 cm. Dimensions: 85, 200, 45.
- Far Right:** 2 ϕ 12 L=80 cm. Dimensions: 45, 45.

Technical drawing of a reinforced concrete slab and column. The drawing includes a plan view of the slab with dimensions (130 cm by 75 cm) and reinforcement details (1Φ14 Pos 1, 2Φ14 Pos 2, 1Φ14 Pos 1, 2Φ16 Pos 3, 2Φ20 Pos 4, 2Φ20 Pos 5). A cross-section A-A' shows the slab thickness and reinforcement. A detail view of the column shows the reinforcement layout. A table of properties is provided: Pt Obj: 1432, Pt Elm: 1432, U1 = 0, U2 = 0, U3 = 0.008, R1 = 0.00104, R2 = 0.00025, R3 = 0.